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Rotatable car seat and base

Abstract

Disclosed herein is a car seat comprised of a base and a shell. The base includes a receiving portion and a backing portion. The seat shell includes a seat back and a seat bottom having a lower portion. The seat bottom is convex and interfaces with the receiving portion of the base. The seat shell is rotatable with respect to the base. The seat shell is selectively positionable between a plurality of orientations. The seat shell includes handles configured to independently unlock rotation of the seat shell with respect to the base. The lower portion of the seat shell includes an aperture configured to receive at least a portion of a projection of the base therethrough.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of and claims priority under 35 U.S.C. § 120 to co-pending U.S. application Ser. No. 16/993,365, filed Aug. 14, 2020, the disclosure of which is fully incorporated by reference as if set forth in its entirety herein.

TECHNICAL FIELD

(1) The present disclosure relates generally to car seats for use in motor vehicles. More particularly, the present disclosure relates to car seats comprised of a base and a mated seat shell attached to the base and rotatable with respect to the base such that the seat shell is selectively positionable between a plurality of positions (e.g., a rearward-facing position, a forward-facing position, an intermediate loading position).

BACKGROUND

(2) In the United States, child car seats are required by law and recommended for children of certain ages or sizes. There are many types of car seats available for purchase. Infant car seats recommended for children under age 1 are typically rear-facing. Many infant car seats include a base that is belted or tethered into the car and a seat that may latch into the base. A popular car seat option for infants as well as children is a “convertible” car seat that may be oriented in a rear-facing position and then “converted” to a front-facing position for accommodating older children.

(3) Conventional convertible car seats are typically designed in such a way that the seat may be uninstalled and reinstalled into the base in order to reposition the seat (e.g., to switch the orientation of the seat or the recline of the seat).

BRIEF SUMMARY

(4) Disclosed herein are car seats. The car seats may be configured to be installed in an associated vehicle. The car seats may be installed in the vehicle using a tether, such as a safety belt of the vehicle (e.g., an existing lap belt of the vehicle) or a LATCH system provided with the car seat. The car seats may be comprised of a base and a seat shell.

(5) The base may include a receiving portion. The base may, in certain constructions, further include a backing portion. The backing portion may extend upwardly away from the receiving portion.

(6) The seat shell may include a seat bottom. The seat bottom may interface with the receiving portion of the base. The seat shell may, in certain constructions, further include a seat back. The seat back may extend upwardly away from the seat bottom. The seat shell may, in certain constructions, include a lower portion. The lower portion may be part of the seat bottom. The lower portion of the seat bottom may interface with the receiving portion of the base.

(7) The seat shell may be removably attached to the base. The seat shell may be rotatable with respect to the base such that the seat shell is selectively positionable between a plurality of positions (e.g., a rearward-facing position, a forward-facing position, an intermediate loading position). In embodiments, the seat shell may specifically be rotatable with respect to the base such that the seat shell is selectively positionable between at least the rearward-facing position and the forward-facing position.

(8) In particular embodiments, the receiving portion of the base may be curved or concave so as to at least partially define a bowl shape. In further embodiments, the lower portion of the seat shell may be curved or convex so as to at least partially define a dome shape.

(9) The receiving portion of the base may, in certain constructions, be shaped complementary to the lower portion of the seat shell so as to retard the ability for debris to enter between the seat shell and the base as the seat shell is rotated with respect to the base.

(10) In certain constructions, the car seat may include a recline mechanism. The recline mechanism may be part of the seat bottom. The recline mechanism may be configured to permit the seat shell to recline with respect to the base such that the seat shell is selectively reclinable between a plurality of positions (e.g., an upright position, a fully-reclined position, an intermediate-reclined position). In embodiments, the seat shell may specifically be reclinable with respect to the base such that the seat shell is selectively positionable between at least the upright position and at least one reclined position (e.g., a fully-reclined position, an intermediate-reclined position).

(11) The recline mechanism may, in certain embodiments, include at least one elongate track. The track may be configured to receive a first rod of the seat shell therethrough. The first rod may be adapted for movement within the track as the seat shell is reclined with respect to the base. The recline mechanism may, in certain embodiments, further define a plurality of openings. Each of the plurality of openings may be configured to selectively receive a second rod of the seat shell therethrough. Each of the plurality of openings may also define a discrete recline position. Each of the plurality of openings may be configured to selectively receive the tube of the seat shell therethrough for selectively positioning the seat shell in a discrete recline position. One of the plurality of openings may define the upright position. Another of the plurality of openings may define the at least one reclined position.

(12) In particular embodiments, a belt path may extend across the base. The belt path may be configured to receive a tether therealong. The belt path may be positioned above the lower portion of the seat shell when the seat shell is in the forward-facing position. The belt path may also be positioned rearward of the lower portion of the seat shell when the seat shell is in the forward-facing position. The seat shell may be rotatable and/or reclinable with respect to the base without interfering with the belt path.

(13) In exemplary embodiments, the car seat may include a projection extending upwardly away from an inner surface of the receiving portion of the base. The car seat may further include an aperture in the lower portion of the seat shell. The aperture may be configured to receive at least a portion of the projection of the base therethrough. In other, alternative embodiments, the car seat may include a projection extending outwardly away from an outer surface of the seat bottom of the seat shell. The car seat may further include an aperture in the receiving portion of the base. The aperture may be configured to receive at least a portion of the projection of the seat shell therethrough.

(14) In certain constructions, the projection extending upwardly away from an inner surface of the receiving portion of the base may include a pedestal. The projection may further include a post extending upwardly away from the pedestal. Further yet, the projection may include a plurality of flanges. The flanges may be spaced apart from one another about the post. The flanges may extend outwardly away from the post. In certain embodiments, the pedestal may be shaped so as to at least partially define a cone shape extending upwardly away from the inner surface of the receiving portion of the base such that any debris that has collected between the seat shell and the base is retarded from substantially encumbering rotation of the seat shell with respect to the base.

(15) In particular embodiments, an outer surface of the seat bottom may include a recessed portion. The recessed portion may be shaped complementary to the pedestal of the projection. The pedestal of the projection may be at least partially received in the recessed portion. The aperture in the lower portion of the seat shell may be shaped complementary to the post and the plurality of flanges of the projection. The plurality of flanges and at least a portion of the post may be received through

the aperture.

(16) In exemplary embodiments, the car seat may include an adjustable assembly mounted for sliding movement relative to the seat shell and selectively positionable with respect to the seat shell. The adjustable assembly may include at least a headrest member. The adjustable assembly may further include a backrest member.

(17) The car seat may, in certain embodiments, be configured to be installed in an associated vehicle using a tether (e.g., an existing safety belt of the associated vehicle, a lower anchor system). The car seat may include a belt lock off arm configured to engage (e.g., clamp and/or tension) the tether. The belt lock off mechanism may be movable between (a) an unlocked position in which the belt lock off arm does not engage (e.g., clamp and/or tension) the tether and (b) a locked position in which the belt lock off arm engages (e.g., clamps and/or tensions) the tether. In embodiments, the belt lock off arm may include an indicator configured to indicate whether the belt lock off arm is in the unlocked position or the locked position.

(18) The base may be configured to lock rotation of the seat shell with respect to the base in the rearward-facing position and the forward-facing position.

(19) In certain constructions, the base may include a first indicator on a first side of the base. The base may further include a second indicator on an opposite, second side of the base. Each of the first and second indicators may be configured to indicate whether rotation of the seat shell is locked with respect to the base. In the same or other constructions, the seat shell may include an indicator configured to indicate whether rotation of the seat shell is locked with respect to the base.

(20) The car seat may also include a handle configured to unlock rotation of the seat shell with respect to the base. In certain constructions, a first handle may be positioned on a first side of the seat shell. A second handle may be provided on an opposite, second side of the seat shell. The seat shell may include a locking projection on the lower portion of the seat shell. The locking projection may be configured to engage with the base so as to lock rotation of the seat shell with respect to the base in the rearward-facing position and the forward-facing position. Each of the first and second handles may be operatively connected to the locking projection. Each of the first and second handles may be configured to independently move the locking projection out of engagement with the base so as to unlock rotation of the seat shell with respect to the base.

(21) In particular embodiments, the car seat may include abutment features. In certain constructions, a first abutment feature may be located on the lower portion of the seat shell proximate a first side thereof. A second abutment feature may be located on the lower portion of the seat shell proximate an opposite, second side thereof. Each of the first and second abutment features may be configured to prevent full 360° rotation of the seat shell with respect to the base when the seat shell is in an intermediate-reclined position.

(22) In exemplary embodiments, the seat shell may include a plurality of engagement features. The base may include a plurality of receiving features. The receiving features of the base may be shaped complementary to the engagement features of the seat shell. Each of the plurality of receiving features of the base may be configured to engage a corresponding one of the plurality of engagement features of the seat shell to guide the seat shell into the rearward-facing position or the forward-facing position. A first one of the plurality of receiving features of the base may be positioned within the receiving portion of the base and may be configured to engage a corresponding first one of the plurality of engagement features of the seat shell that is positioned on the lower portion of the seat shell. A second one of the plurality of receiving features of the base may be positioned adjacent the receiving portion of the base and may be configured to engage a corresponding second one of the plurality of engagement features of the seat shell that is positioned adjacent the lower portion of the seat shell.

(23) In particular embodiments, the seat shell may further be selectively positionable in an intermediate position between the rearward-facing position and the forward-facing position. The intermediate position may, in certain embodiments, be a predetermined position between the

rearward-facing position and the forward-facing position. In such embodiments, the predetermined intermediate position may be defined by one or more detents and/or one or more engagement features. In such embodiments, there are a finite number of intermediate positions into which the seat shell may be selectively positioned. In other embodiments, the intermediate position may be any position between the rearward-facing position and the forward-facing position such that the seat shell is freely positionable in the intermediate position. In such embodiments, there are an infinite number of intermediate positions into which the seat shell may be selectively positioned. In certain constructions, the base may include a first retainment feature on a first side of the base. The base may also include a second retainment feature on an opposite, second side of the base. The seat shell may include a first engagement feature at a forward end of the seat shell. The seat shell may further include a second engagement feature at an opposite, rear end of the seat shell. Each of the first and second retainment features of the base may be configured to engage a corresponding one of the first and second engagement features of the seat shell to retain the seat shell in the intermediate position. The first and second retainment locking features of the base may be formed as nubs or projections. The first and second engagement features of the seat shell may include an aperture or detent configured to at least partially receive a corresponding one of the nubs or projections.

(24) In certain constructions, the car seat may include an upper tethering point proximate an upper edge of the backing portion of the base. The upper tethering point may be configured to remain stationary as the seat shell is rotated with respect to the base.

(25) In accordance with the foregoing, disclosed in exemplary embodiments herein are car seats. In accordance with particular embodiments of the present disclosure, an exemplary car seat comprises: a base including a receiving portion; a seat shell including a seat bottom having a lower portion and a seat back extending upwardly away from the seat bottom, the seat bottom interfacing with the receiving portion of the base, wherein the seat shell is removably attached to the base and rotatable with respect to the base such that the seat shell is selectively positionable between at least a rearward-facing position and a forward-facing position; and a recline mechanism in the seat bottom, the recline mechanism configured to permit the seat shell to recline with respect to the base such that the seat shell is selectively reclinable between at least an upright position and at least one reclined position.

(26) In accordance with other particular embodiments of the present disclosure, an exemplary car seat comprises: a base including a receiving portion; a seat shell including a seat bottom having a lower portion and a seat back extending upwardly away from the seat bottom, the seat bottom interfacing with the receiving portion of the base, wherein the seat shell is removably attached to the base and rotatable with respect to the base such that the seat shell is selectively positionable between at least a rearward-facing position and a forward-facing position; a locking projection on the lower portion of the seat shell and configured to engage the base to lock rotation of the seat shell with respect to the base in each of the rearward-facing position and the forward-facing position; and a first handle positioned on a first side of the seat shell and a second handle positioned on an opposite, second side of the seat shell, each of the first and second handles operatively connected to the locking projection and configured to independently move the locking projection out of engagement with the base so as to unlock rotation of the seat shell with respect to the base.

(27) In accordance with other particular embodiments of the present disclosure, an exemplary car seat is configured to be installed in an associated vehicle using a tether (e.g., an existing safety belt of the associated vehicle, a lower anchor system). The car seat may comprise: a base including a receiving portion; a seat shell including a seat bottom having a lower portion and a seat back extending upwardly away from the seat bottom, the seat bottom interfacing with the receiving portion of the base, wherein the seat shell is removably attached to the base and rotatable with respect to the base such that the seat shell is selectively positionable between at least a rearward-

facing position and a forward-facing position; a projection extending upwardly away from an inner surface of the receiving portion of the base; and an aperture in the lower portion of the seat shell, the aperture configured to receive at least a portion of the projection therethrough.

(28) Other aspects of the invention may be apparent to those of ordinary skill with review of the attached drawings, appended claims, and additional description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The foregoing and other features of the present disclosure will become more fully apparent from the following description and appended claims, taken in conjunction with the accompanying drawings. Understanding that these drawings depict several examples in accordance with the disclosure and are, therefore, not to be considered limiting of its scope, the disclosure is described with additional specificity and detail below through the use of the accompanying drawings.

(2) FIG. 1 is an exploded view of a car seat in accordance with the disclosure.

(3) FIG. 2 is a perspective view of the car seat of FIG. 1 in a forward-facing position.

(4) FIG. 3 is a side view of the car seat of FIG. 1 in the forward-facing position.

(5) FIG. 4 is a perspective view of the car seat of FIG. 1 in a rearward-facing position.

(6) FIG. 5 is a side view of the car seat of FIG. 1 in the rearward-facing position.

(7) FIG. 6 is a perspective view of the car seat of FIG. 1 in an intermediate position.

(8) FIG. 7 is a side view of the car seat of FIG. 1 in the intermediate position.

(9) FIG. 8 is a perspective view of the base of the car seat of FIG. 1.

(10) FIG. 9A is a first side view of the base of FIG. 8.

(11) FIG. 9B is a second side view of the base of FIG. 8.

(12) FIG. 10 is a front view of the base of FIG. 8.

(13) FIG. 11 is a rear view of the base of FIG. 8.

(14) FIG. 12 is a top view of the base of FIG. 8.

(15) FIG. 13 is a perspective view of the receiving portion of the base of FIG. 8.

(16) FIG. 14 is a perspective view of the seat shell of the car seat of FIG. 1.

(17) FIG. 15A is a first side view of the seat shell of FIG. 14.

(18) FIG. 15B is a second side view of the seat shell of FIG. 14.

(19) FIG. 16 is a front view of the seat shell of FIG. 14.

(20) FIG. 17 is a bottom view of the seat shell of FIG. 14.

(21) FIG. 18 is a perspective view of the lower portion of the seat shell of FIG. 14.

(22) FIG. 19 is a perspective view of the forward end of the lower portion of the seat shell of FIG. 14.

(23) FIG. 20 is a perspective view of the rear end of the lower portion of the seat shell of FIG. 14.

(24) FIG. 21 is a first side cross-sectional view taken along line 21-21' of FIG. 16 showing features of the recline mechanism.

(25) FIG. 22 is a first side perspective cutaway view of the lower portion of the seat shell of FIG. 14 showing features of the recline mechanism.

(26) FIG. 23 is a second side perspective cutaway view of the lower portion of the seat shell of FIG. 14 showing features of the recline mechanism.

DETAILED DESCRIPTION

(27) In the following detailed description, reference is made to the accompanying drawings, which form a part hereof. In the drawings, similar symbols identify similar components, unless context dictates otherwise. The illustrative examples described in the detailed description and drawings are not meant to be limiting and are for explanatory purposes. Other examples may be utilized, and other changes may be made, without departing from the spirit or scope of the subject matter

presented herein. It will be readily understood that the aspects of the present disclosure, as generally described herein and illustrated in the drawings, may be arranged, substituted, combined, and designed in a wide variety of different configurations, each of which are explicitly contemplated and form a part of this disclosure.

(28) It should be noted that some of the terms used herein may be relative terms. For example, the terms “upper” and “lower” and the terms “forward” and “rearward” or are relative to each other in location, i.e. an upper component is located at a higher elevation than a lower component in a given orientation, but these terms may change if the device is flipped. An intermediate component, on the other hand, is always located between an upper component and a lower component regardless of orientation. The terms “horizontal” and “vertical” are used to indicate direction relative to an absolute reference, i.e. ground level. However, these terms should not be construed to require structures to be absolutely parallel or absolutely perpendicular to each other. For example, a first vertical structure and a second vertical structure are not necessarily parallel to each other. The terms “top” and “bottom” or “base” are used to refer to surfaces where the top is always higher than the bottom/base relative to an absolute reference, i.e. the surface of the earth. The terms “upwards” or “upwardly” and “downwards” or “downwardly” are also relative to an absolute reference; upwards is always against the gravity of the earth. The terms “forward” and “rearward” or “rear” with respect to a position or orientation are opposite one another along a common direction, and an “intermediate” position is always located between a forward position and a rearward position. The term “parallel” should be construed in its lay sense of two surfaces that maintain a generally constant distance between them, and not in the strict mathematical sense that such surfaces will never intersect when extended to infinity. The terms “operative to” and “adapted to” and similar terms are used herein to describe that a particular component has certain structural features designed to perform a designated function. Such components should be construed as having the expressed structure, with the designated function being considered part of the structure. The term “engage” and similar terms are used herein to describe the interaction between particular components and does not necessarily require that such components contact one another (directly or indirectly). As used herein and as will be appreciated by those skilled in the art, the term “car seat” encompasses car seats, safety seats, restraints, boosters, and the like for children, infants, and toddlers and the like.

(29) In accordance with aspects of the disclosure, exemplary embodiments of car seats are illustrated in various levels of specificity in FIGS. 1-20. As will be appreciated by those skilled in the art, the car seats described herein may be convertible car seats that are convertible between a forward-facing configuration, a rearward-facing configuration, and a booster configuration. The car seats of the present disclosure may be configured to support an infant, child, toddler, or the like. Very generally, car seat **10** is designed for safely transporting an infant or child in a car. As illustrated in FIG. 1, the car seat **10** generally includes a seat shell **200**, which is supported on a base **300**. As will be readily understood by those skilled in the art, the car seat **10** (and, namely, the base **300**) may be affixed to the seat of a car, such as by a seat belt (e.g., lap belt) of the car.

(30) As described in more detail herein, the seat shell **200** may be removably attached to the base **300**. Additionally, the seat shell **200** may be rotatable with respect to the base **300**. The seat shell **200** may be rotated to be selectively positionable in a variety of different positions (e.g., e.g., a rearward-facing position, a forward-facing position, an intermediate loading position).

(31) FIG. 2 and FIG. 3 depict the seat shell **200** supported on the base **300** in a forward-facing position. With reference to FIG. 3, in the forward-facing position, a forward end **206** of the seat shell **200** and a forward end **306** of the base **300** face a common direction (e.g., toward the forward end of the vehicle when installed), and a rear end **208** of the seat shell **200** and a rear end **308** of the base **300** face a common direction (e.g., toward the rear end of the vehicle when installed).

(32) FIG. 4 and FIG. 5 depict the seat shell **200** supported on the base **300** in a rearward-facing position. As can be readily understood with reference back to FIG. 2 and FIG. 3, the seat shell **200**

is rotated with respect to the base **300** about 180° between the forward-facing position (FIG. **2** and FIG. **3**) and the rearward-facing position (FIG. **4** and FIG. **5**). Put another way, the seat shell **200** generally faces one direction in the forward-facing position and an opposite direction in the rearward-facing position. With reference to FIG. **5**, in the rearward-facing position, the rear end **208** of the seat shell **200** and the forward end **306** of the base **300** face a common direction (e.g., toward the forward end of the vehicle when installed), and the forward end **206** of the seat shell **200** and the rear end **308** of the base **300** face a common direction (e.g., toward the rear end of the vehicle when installed).

(33) FIG. **6** and FIG. **7** depict the seat shell **200** supported on the base **300** in an intermediate position. As can be readily understood with reference back to FIG. **2** and FIG. **3** and FIG. **4** and FIG. **5**, the seat shell **200** is rotated with respect to the base **300** about 90° between either the forward-facing position (FIG. **2** and FIG. **3**) or the rearward-facing position (FIG. **4** and FIG. **5**) and the intermediate position (FIG. **6** and FIG. **7**). Put another way, the seat shell **200** generally faces one direction in the forward-facing position, an opposite direction in the rearward-facing position, and a direction in between when in the intermediate position. With reference to FIG. **7**, in the intermediate position, the forward end **206** of the seat shell **200** faces a direction between the forward end **306** of the base **300** and the rear end **308** of the base **300** (e.g., such that, when installed, the forward end **306** of the seat shell is toward a side door between the forward and rear ends of the vehicle), and the rear end **208** of the seat shell **200** faces a direction between the forward end **306** of the base **300** and the rear end **308** of the base **300** (e.g., such that, when installed, the rear end **308** of the seat shell is away from the side door between the forward and rear ends of the vehicle when installed).

(34) As may now be appreciated, the seat shell **200** may generally be rotated with respect to the base **300** between the forward-facing position and the rearward-facing position as the base **300** remains installed and stationary in the vehicle. Further yet, the seat shell **200** may generally be rotated any desired amount with respect to the base **300** (including being rotated a full 360°). As may further be appreciated, the seat shell **200** is rotatable with respect to the base **300** to a variety of intermediate positions between the forward-facing position and the rearward-facing position. In this regard, it is specifically contemplated that the seat shell **200** may be rotatable with respect to the base **300** about 90° in both a clockwise direction and a counterclockwise direction to intermediate positions. In this way, the car seat **10** is adapted to be usable on either the driver's side or passenger's side of a vehicle and equally provides ease for a user to place a child into the car seat from a side door of the vehicle with the seat shell **200** generally facing the user. For example, in the intermediate position illustrated in FIG. **6** and FIG. **7**, the seat shell **200** generally faces the passenger's side door of the vehicle and advantageously enables the user to quickly and easily place or remove a child from the car seat while facing the child.

(35) FIGS. **8-13** illustrate various views of the base **300**. Very generally, the base **300** may support and stabilize the seat shell **200**. The base **300** may include a first side **302** and a second side **304** opposite the first side **302**. The base **300** may further include a forward end **306** and a rear end **308** opposite the forward end **306**.

(36) In particular, the base **300** may include a receiving portion **310** and a backing portion **350**. The receiving portion **310** of the base **300** may generally operate to support the seat shell **200** for rotation with respect thereto. For example, at least a portion of the seat shell **200** may be received by and/or within the receiving portion **310** of the base **300**. Put another way, the receiving portion **310** of the base **300** may be configured to interface with at least a portion of the seat shell **200**. As can be best seen in FIGS. **8-10**, the backing portion **350** may extend upwardly away (i.e., in the direction of arrow **50**) from the receiving portion **310**, such as extending upwardly from the receiving portion **310** proximate the rear end **308** of the base **300**.

(37) With particular reference to FIG. **8**, in this exemplary embodiment, the receiving portion **310** of the base **300** is curved or concave so as to at least partially define a bowl shape. In this way, the

receiving portion **310** of the base **300** generally defines a cavity **316** that is sized and shaped to receive at least a portion of the seat shell **200**.

(38) As described in detail herein, the base **300** is configured to lock rotation of the seat shell **200** with respect to the base **300**. In particular, the base **300** is configured to lock rotation of the seat shell **200** with respect to the base **300** in each of the forward-facing position (refer to FIG. 2 and FIG. 3) and the rearward-facing position (refer to FIG. 4 and FIG. 5).

(39) In certain constructions, the base **300** may include one or more indicators configured to indicate whether rotation of the seat shell **200** is locked with respect to the base **300**. As best seen in FIG. 9A, a first indicator **303** may be positioned on the first side **302** of the base **300**. Similarly, as best seen in FIG. 9B, a second indicator **305** may be positioned on the opposite, second side **304** of the base **300**. Each of the first and second indicators **303**, **305** may be configured to indicate whether rotation of the seat shell **200** is locked with respect to the base **300** (e.g., particularly to indicate whether rotation of the seat shell **200** is locked in the forward-facing position or the rearward-facing position). In particular, the first and second indicators **303**, **305** may be configured to visibly indicate whether rotation of the seat shell **200** is locked with respect to the base **300**. By way of non-limiting example, the first and second indicators **303**, **305** may be configured to give a visual indication in a first color (e.g., red) when rotation of the seat shell **200** is unlocked with respect to the base **300** and/or when the seat shell **200** is not in the forward-facing position or the rearward-facing position, and the first and second indicators **303**, **305** may be configured to give a visual indication in a second, different color (e.g., green) when rotation of the seat shell **200** is locked with respect to the base **300** and/or when the seat shell **200** is in the forward-facing position or the rearward-facing position.

(40) With continued reference to FIG. 9A and FIG. 9B, the base **300** may include a belt path **312**. The belt path **312** may be configured to receive a tether therealong (e.g., along a length of the base **300**). In some embodiments, the tether can be an existing safety belt of a vehicle. In other embodiments, the tether can be any other form of tether (e.g., a tether provided with the car seat, a lower anchor system). The belt path **312** may generally extend across the base **300**. For example, it can be seen in FIGS. 9-11 that the belt path **12** generally extends from the first side **302** of the base **300** to the second side **304** of the base **300**. With specific comparison between FIG. 3 (depicting the seat shell **200** supported on the base **300** in the forward-facing position) and FIG. 9A, it can be seen that the belt path **312** is positioned above and rearward of a lower portion **210** of the seat shell **200**. This advantageously allows the seat shell **200** to be rotated and/or reclined with respect to the base **300** without interfering with the belt path **312** and without any need to uninstall the base **300** or otherwise reroute or reposition the tether when it is desired to rotate the seat shell **200** with respect to the base **300**. In addition, the seat shell **200** is readily removable or detachable from the base **300** without any need to uninstall or reinstall the base **300** or otherwise reroute or reposition the tether.

(41) With reference now to FIG. 12, the backing portion **350** of the base **300** may include an upper tethering point **356**. The upper tethering point **356** is generally designed to operate as a tether anchor. The upper tethering point **356** may be positioned proximate an upper edge **352** of the backing portion **350** of the base **300**. The upper tethering point **356** may be configured to remain stationary as the seat shell **200** is rotated with respect to the base **300**. This advantageously allows the seat shell **200** to be rotated and/or reclined with respect to the base **300** without interfering with the upper tethering point **356** and without any need to uninstall the base **300** or otherwise reroute or reposition a tether attached to the upper tethering point **356** when it is desired to rotate the seat shell **200** with respect to the base **300**. In addition, the seat shell **200** is readily removable or detachable from the base **300** without any need to uninstall or reinstall the base **300** or otherwise reroute or reposition a tether attached to the upper tethering point **356**. As such, the car seat may advantageously remain constantly anchored via the upper tethering point **356** without interfering with rotation and/or reclination of the seat shell **200** with respect to the base **300**.

(42) With continued reference to FIG. 12, the cavity **316** of the receiving portion **310** of the base

300 can be more clearly seen. The cavity **316** generally has an open or missing top face and is at least partially defined and bound by an inner surface **314** of the receiving portion **310** of the base **300**. As can be seen in FIG. 12 and FIG. 13, the receiving portion **310** of the base **300** may include a projection **320**. The projection **320** may extend upwardly away from the inner surface **314** of the receiving portion **310** of the base **300**, such as extending upwardly from the inner surface **314** of the receiving portion **310** proximate a midpoint between the first side **302** and the second side **304** of the base **300**. That is, the projection **320** may be positioned substantially centrally between the first side **302** and the second side **304** of the base **300**. As can be seen in FIG. 12, the projection **320** may generally be positioned closer to the rear end **308** of the base **300** than the front end **306** of the base **300**. As can be further seen in FIG. 12, the projection **320** may generally be angled toward the front end **306** of the base **300**. The angling of the projection **320** may help ensure that as the seat shell **200** is rotated with respect to the base **300**, the seat shell **200** remains engaged with and correctly angled (i.e., reclined) with respect to the base **300**, including in both the rearward-facing and forward-facing positions.

(43) The receiving portion **310** of the base **300** may include a support ledge **358**. The support ledge **358** may generally operate as a surface along which at least a portion of the seat shell **200** travels as the seat shell **200** is rotated with respect to the base **300**, as described in detail herein. The support ledge **358** may be positioned radially inward of an outer edge of the receiving portion **310** of the base **300**. As seen in FIG. 12 and FIG. 13, the support ledge **358** may generally be circular in shape and define an opening to the cavity **316** of the receiving portion **310** of the base **300**.

(44) The receiving portion **310** of the base **300** may include one or more retainment features. For example, as shown in FIG. 12 and FIG. 13, a first retainment feature **342** may be positioned on the first side **302** of the base **300**. A second retainment feature **344** may be positioned on the opposite, second side **304** of the base **300**. As best seen in FIG. 12 and FIG. 13, the first and second retainment features **342**, **344** may be positioned on the support ledge **358**. The first and second retainment features **342**, **344** of the base **300** generally operate so as to retain the seat shell **200** in the intermediate position (refer to FIG. 6 and FIG. 7). More specifically, as described in detail herein, each of the first and second retainment features **342**, **344** may be configured to engage at least a portion of the seat shell **200** to retain the seat shell **200** in the intermediate position. By way of non-limiting example, the first and second retainment features **342**, **344** may be formed as nubs or projections. Alternatively, the seat shell **200** could be formed with corresponding nubs or projections and the first and second retainment features **342**, **344** could be formed as detents or apertures configured to engage with the corresponding nubs or projections of the seat shell **200**.

(45) With continued reference to FIG. 12, the base **300** may include one or more receiving features. For example, as shown in FIG. 12 and FIG. 13, an upper receiving feature **330** may be positioned on the backing portion **350** of the base **300**. More specifically, the upper receiving feature **330** may be positioned on the backing portion **350** of the base **300** proximate the upper edge **352** of the backing portion **350** of the base **300**. By way of non-limiting example, as may be best seen with reference back to FIG. 8 and FIGS. 9A-B, the upper receiving feature **330** may be formed as a hook. In particular, the upper receiving feature **330** may be formed as a downward-pointing or downward-oriented hook (i.e., in the direction of arrow **60** in FIG. 8 and FIGS. 9A-B). In certain constructions and as best seen in FIG. 12, the upper receiving feature **330** may also define the upper tethering point **356**.

(46) As shown in FIG. 12 and FIG. 13, an intermediate receiving feature **331** may be positioned proximate the rear end **308** of the base **300**. In particular, the intermediate receiving feature **331** may be positioned proximate a location where the backing portion **350** of the base **300** meets the receiving portion **310** of the base **300**. More specifically, the intermediate receiving feature **331** may be positioned proximate the support ledge **358** of the receiving portion **310** of the base **300**. By way of non-limiting example, the intermediate receiving feature **331** may be formed as a hook. In particular, the intermediate receiving feature **331** may be formed as a downward-pointing or

downward-oriented hook.

(47) With continued reference to FIG. 12 and FIG. 13, a lower receiving feature 332 may be positioned within the receiving portion 310 of the base 300. In particular, the lower receiving feature 332 may be positioned within the receiving portion 310 of the base 300 proximate the rear end 308 of the base 300. More specifically, the lower receiving feature 332 may be positioned within the receiving portion 310 of the base 300 on one side of the belt path 312. By way of non-limiting example, the lower receiving feature 332 may be formed as a hook. In particular, the lower receiving feature 332 may be formed as a downward-pointing or downward-oriented hook.

(48) The upper, intermediate, and lower receiving features 330, 331, 332 of the base 300 generally operate so as to retain the seat shell 200 in the forward-facing position (refer to FIG. 2 and FIG. 3) or the rearward-facing position (refer to FIG. 4 and FIG. 5). More specifically, as described in detail herein, each of the upper, intermediate, and lower receiving features 330, 331, 332 may be configured to engage at least a portion of the seat shell 200 to retain the seat shell 200 in the forward-facing position or the rearward-facing position.

(49) As can be best seen in the exemplary embodiment illustrated in FIG. 12, the base 300 may include a belt lock off arm 500. The belt lock off arm 500 can be connected to the base 300. A proximal end of belt lock off arm 500 may be connected to the base 300 on one side of the belt path 312. In particular, as shown in FIG. 12, the belt lock off arm 500 may be connected to the base 300 proximate the rear end 308 of the base 300. In this way, the belt lock off arm 500 may be configured to extend across the belt path 312. A distal end of the belt lock off arm 500 may be spaced apart from the proximal end of the belt lock off arm 500. The belt lock off arm 500 may be pivotally movable with respect to the base 300 such that the distal end of the belt lock off arm 500 is capable of moving toward and away from the base 300 (and, more particularly, toward and away from the belt path 312) between a locked position and an unlocked position. In certain non-depicted embodiments, the belt lock off arm 500 may also be slidably movable with respect to the base 300 such that the belt lock off arm 500 is slidable into and out of the base 300. The belt lock off arm 500 may be slidable into and out of the base 300 in a direction substantially perpendicular to the belt path 312.

(50) The belt lock off arm 500 may generally operate so as to engage with (e.g., clamp, tension) a tether extending along the belt path 312 in a locked position (e.g., as shown in FIG. 12). In embodiments, the belt lock off arm 500 is configured to clamp the tether. In other embodiments, the belt lock off arm 500 is configured to tension the tether. In further embodiments, the belt lock off arm 500 is configured to clamp and tension the tether. More specifically, the belt lock off arm 500 may be movable between (a) an unlocked position in which the belt lock off arm 500 is not configured to engage (e.g., clamp and/or tension) the tether of the associated vehicle, and (b) a locked position in which the belt lock off arm 500 is configured to engage (e.g., clamp and/or tension) the tether of the associated vehicle. As will be appreciated by those skilled in the art, the belt lock off arm 500 may increase the mechanical advantage in the system by using leverage to assist in clamping and/or tensioning the tether. This advantageously decreases the amount of force a user would otherwise be required to apply to clamp and/or tension the tether and likewise advantageously decreases the potential for an improper, unsafe, or otherwise undesirable installation.

(51) In certain constructions, the belt lock off arm 500 may include one or more indicators configured to indicate whether the belt lock off arm 500 is in the locked position. As best seen in FIG. 12, an indicator 502 may be positioned on an upper face of the belt lock off arm 500. The indicator 502 may be configured to indicate the belt lock off arm 500 is in the locked position in which the belt lock off arm 500 is configured to engage (e.g., clamp and/or tension) the tether of the associated vehicle. In particular, the indicator 502 may be configured to visibly indicate whether the belt lock off arm 500 is in the locked position. By way of non-limiting example, the indicator 502 may be configured to give a visual indication in a first color (e.g., red) when the belt

lock off arm **500** is in the unlocked position and another visual indication in a second, different color (e.g., green) when the belt lock off arm **500** is in the locked position.

(52) As can be best seen in FIG. **13**, the belt lock off arm **500** may be designed to be generally U-shaped, although embodiments of the present disclosure are not so limited and the belt lock off arm **500** may be of any suitable size and shape. In this regard, when the belt lock off arm **500** is in the locked position (refer to FIG. **12** and FIG. **13**), the belt lock off arm **500** may lie about (e.g., around) the projection **320**. More specifically, in certain constructions, the belt lock off arm **500** may lie within a recessed portion in the inner surface **314** of the receiving portion of the base **300**. In this way, the projection **320** may extend upwardly away from the inner surface **314** of the receiving portion of the base **300** above the belt lock off arm **500**, such that the belt lock off arm **500** does not interfere with rotation and/or reclination of the seat shell **200** with respect to the base **300** as described in detail herein. Additionally, in the locked position, the belt lock off arm **500** may lie about the lower receiving feature **332**.

(53) As can be best seen in the exemplary embodiment illustrated in FIG. **13**, the projection **320** may include a pedestal **322**. The pedestal **322** of the projection **320** may extend upwardly from the inner surface **314** of the receiving portion **310** of the base **300**. In this way, the pedestal **322** of the projection **320** may be shaped so as to at least partially define a cone shape (i.e., by tapering inwardly in diameter as the pedestal **322** of the projection **320** extends away from the inner surface **314** of the receiving portion **310** of the base **300**). In this way, the pedestal **322** of the projection **320** may generally extend upwardly away from the inner surface **314** of the receiving portion **310** of the base **300**. As a result, any debris that has collected between the seat shell **200** and the base **300** may generally be prevented from interfering with the interface between the seat shell **200** and the base **300**. In this way, any debris that accumulates between seat shell **200** and the base **300** may be prevented or retarded from substantially encumbering rotation of the seat shell **200** with respect to the base **300**. The projection **320** may further include a post **324**. The post **324** of the projection **320** may extend upwardly from the pedestal **322** of the projection **320**. The projection **320** may further include one or more flanges **326**. In the exemplary embodiment illustrated in FIG. **13**, the projection **320** includes a plurality of flanges **326**. The flanges **326** of the projection **320** may be spaced apart from one another about the post **324** of the projection **320**. The flanges **326** of the projection **320** may extend outwardly away from the post **324** of the projection **320**. In certain constructions, the flanges **326** of the projection **320** may extend outwardly away from the post **324** of the projection **320** on opposite sides of the post **324**. In particular embodiments, a first one of the flanges **326** of the projection **320** may extend outwardly away from the post **324** of the projection **320** toward the first side **302** of the base **300**, and a second one of the flanges **326** of the projection **320** may extend outwardly away from the post **324** of the projection **320** toward the second side **304** of the base **300**. The flanges **326** of the projection **320** may be spaced apart from the pedestal **322** of the projection **320**.

(54) FIGS. **14-20** illustrate various views of the seat shell **200**. Very generally, the seat shell **200** may support and stabilize an occupant (e.g., an infant or child) therein and/or thereon. The seat shell **200** may include a first side **202** and a second side **204** opposite the first side **202**. The seat shell **200** may further include a forward end **206** and a rear end **208** opposite the forward end **206**. As described in detail herein, the seat shell **200** may be supported on the base **300**, and the seat shell **200** may be rotatable with respect to the base **300** such that the seat shell **200** is selectively positionable in a variety of different positions (e.g., a rearward-facing position, a forward-facing position, an intermediate loading position). In particular, the seat shell **200** is selectively positionable in at least a forward-facing position (refer to FIG. **2** and FIG. **3**) and a rearward-facing position (refer to FIG. **4** and FIG. **5**). As described in detail herein, the seat shell **200** may also be removably attached to the base **300**.

(55) In particular, the seat shell **200** may include a seat bottom **209**, a seat back **250**, and a lower portion **210**. The seat bottom **209** of the seat shell **200** may generally operate as a seating surface

for the occupant by supporting and stabilizing the butt of the occupant. For example, an occupant may be seated upon a top surface of the seat bottom **209** of the shell. As can be best seen in FIG. **14** and FIGS. **15A-B**, the seat back **250** may extend upwardly away (i.e., in the direction of arrow **50**) from the seat bottom **209**, such as extending upwardly from the seat portion **209** proximate the rear end **208** of the seat shell **200**. The seat bottom **209** may interface with the receiving portion **310** of the base.

(56) The seat shell **200** may include a lower portion **210**. The lower portion **210** may, in certain embodiments, be part of the seat bottom **209** and may be integrally formed therewith. As described in detail herein, the lower portion **210** may interface with the receiving portion **310** of the base **300**. As described in detail herein, the receiving portion **310** of the base **300** may generally operate to support the seat shell **200** for rotation with respect thereto. For example, the lower portion **210** of the seat shell **200** may be received by and/or within the receiving portion **310** of the base **300**. Put another way, the lower portion **210** of the seat shell **200** may be configured to interface with the receiving portion **310** of the base **300**. In particular, the cavity **316** of the receiving portion **310** of the base **300** may receive the lower portion **210** of the seat shell **200**.

(57) With particular reference to FIG. **9A** and FIG. **9B**, in this exemplary embodiment, the lower portion **210** of the seat shell **200** is curved or convex so as to at least partially define a dome shape. In this way, the lower portion **210** of the seat shell **200** may be shaped complementary to the concave, bowl-shaped receiving portion **310** of the base **300**. Advantageously, the complementary shapes of the lower portion **210** of the seat shell **200** and the receiving portion **310** of the base **300** assists in retarding the ability for debris to enter between the seat shell **200** and the base **300** as the seat shell **200** is rotated with respect to the base **300**. Constructing the car seat **10** such that debris is retarded from entering between the seat shell **200** and the base **300** (e.g., such that minimal or no debris gaps are defined between the seat shell **200** and the base **300**) ensures that rotation of the seat shell **200** with respect to the base **300** can occur substantially unencumbered and without being substantially impeded or retarded by debris that could otherwise accumulate between the seat shell **200** and the base **300**. To assist in preventing debris from entering the receiving portion **310** of the base **300** and/or removing debris, the base **300** may include one or more openings or sets of openings **309**. In certain embodiments, such openings **309** may be defined in the support ledge **358** and/or proximate the support ledge **358**, as can be best seen in FIG. **12**. In other, non-depicted embodiments, the base **300** may include one or more other mechanisms (e.g., wipers) configured to assist in the removal of debris from the receiving portion **310** of the base **300**, namely any debris that has accumulated near the projection **320**.

(58) As described in detail herein, the base **300** is configured to lock rotation of the seat shell **200** with respect to the base **300**. In particular, the base **300** is configured to lock rotation of the seat shell **200** with respect to the base **300** in each of the forward-facing position (refer to FIG. **2** and FIG. **3**) and the rearward-facing position (refer to FIG. **4** and FIG. **5**).

(59) With reference now to FIG. **15A** and FIG. **15B**, the lower portion **210** of the seat shell **200** may include a locking projection **238**. The locking projection **238** may extend outwardly away from the seat shell **200**, such as extending outwardly away from the rear end **208** of the seat shell **200** proximate a location where the seat back **250** of the seat shell **200** meets the lower portion **210** of the seat shell **200**. With reference to FIG. **18**, the locking projection **238** may be positioned substantially centrally between the first side **202** and the second side **204** of the seat shell **200**. As described in detail herein, the locking projection **238** may be configured to be deflectable or movable between a locked position and an unlocked position. The locking projection **238** may be configured to be deflectable or movable into the lower portion **210** of the seat shell **200**. In particular, the locking projection **238** is configured to deflect or move into and out of engagement with the base **300** so as to selectively lock and unlock rotation of the seat shell **200** with respect to the base **300**. More specifically, the locking projection **238** may be biased (e.g., spring-biased) outwardly into the locked position such that the locking projection **238** is configured to

automatically lock rotation of the seat shell **200** with respect to the base **300** when the seat shell **200** is in the forward-facing position (refer to FIG. 2 and FIG. 3) or the rearward-facing position (refer to FIG. 4 and FIG. 5). The locking projection **238** is generally configured such that it does not lock rotation of the seat shell **200** with respect to the base **300** when the seat shell **200** is in the intermediate position (refer to FIG. 6 and FIG. 7).

(60) With continued reference to FIGS. 15A-B and FIG. 16, the seat shell **200** may include one or more actuators or handles. In the exemplary embodiment illustrated in FIG. 15A and FIG. 15B, the seat shell **200** includes a first actuator or handle **262** positioned on the first side **202** of the seat shell **200** and a second actuator or handle **264** positioned on the opposite, second side **204** of the seat shell **200**. More specifically, in this embodiment, the first and second handles **262**, **264** are each positioned proximate an upper edge **252** of the seat back **250** of the seat shell **200**. The first and second handles **262**, **264** are each operatively connected to the locking projection **238**. The first and second handles **262**, **264** generally operate so as to deflect or move the locking projection **238** out of engagement with the base **300** so as to unlock rotation of the seat shell **200** with respect to the base **300**. In particular, the first and second handles **262**, **264** may be configured to independently move the locking projection **238** out of engagement with the base **300** so as to unlock rotation of the seat shell **200** with respect to the base **300**. That is, in certain constructions, actuation of just one of the first or second handles **262**, **264** is sufficient to overcome the biasing force acting upon the locking projection **238** and to move the locking projection **238** out of engagement with the base **300** so as to unlock rotation of the seat shell **200** with respect to the base **300**. In this way, the car seat **10** is adapted to be usable on either the driver's side or passenger's side of a vehicle and equally provides ease for a user to actuate one of the first or second handles **262**, **264** from a side door of the vehicle. For example, this advantageously enables the user to quickly and easily actuate one of the first or second handles **262**, **264** without having to reach across the car seat or an occupant in the car seat.

(61) As seen in FIG. 16, the car seat **10** may include an adjustable assembly **400**. The adjustable assembly **400** may include at least a headrest member. The adjustable assembly **400** may be mounted for sliding movement relative to the seat shell **200**. The adjustable assembly **400** may be selectively positionable with respect to the seat shell **200** (e.g., in a reclined headrest position). Additional features of the adjustable assembly are disclosed in U.S. patent application Ser. No. 15/837,231, entitled Adjustable Headrest for Child Car Seat, the subject matter of which is incorporated herein by reference.

(62) Turning now to FIG. 17 and FIG. 18, the lower portion **210** of the seat shell **200** may include an aperture **220**. The aperture **220** may extend through the outer surface **216** of the lower portion **210** of the seat shell **200**. The aperture **220** may be configured to receive at least a portion of the projection **320** of the base **300** therethrough. In this way, the lower portion **210** of the seat shell **200** may interface with the receiving portion **310** of the base **300**, namely with the aperture **220** in the lower portion **210** of the seat shell **200** interfacing with the projection **320** in the receiving portion **310** of the base **300**. As may now be appreciated, the base **300** may rotatably support the seat shell **200** with the outer surface **216** of the lower portion **210** of the seat shell **200** generally juxtaposed the inner surface **314** of the receiving portion **310** of the base **300**. As may now be appreciated, the seat bottom **209** (namely the lower portion **210** thereof) of the seat shell **200** may interface with the receiving portion **310** of the base **300** via the aperture **220** in the lower portion **210** of the seat shell **200** interfacing with the projection **320** in the receiving portion **310** of the base **300**. In alternative embodiments, it is contemplated that the interfacing structures of the seat bottom **209** of the seat shell **200** and the receiving portion **310** of the base **300** could be reversed. That is, it is specifically contemplated that in certain non-depicted embodiments, the receiving portion **310** of the base **300** could be formed with an aperture that interfaces with a projection of the seat bottom **209** of the seat shell **200**.

(63) As can be best seen in the exemplary embodiment illustrated in FIG. 17 and FIG. 18, the outer

surface **216** of the lower portion **210** of the seat shell **200** may include a recessed portion **222**. The recessed portion **222** may extend inwardly from the outer surface **216** of the lower portion **210** of the seat shell **200**. As may be best seen in FIG. **18** with reference also to FIG. **13**, the recessed portion **222** of the outer surface **216** of the lower portion **210** of the seat shell **200** may be shaped complementary to the pedestal **322** of the projection **320** of the base **300**. In this way, the pedestal **322** of the projection **320** may be at least partially received within the recessed portion **222**. Put another way, at least a portion of the pedestal **322** of the projection **320** may be received within the recessed portion **222**. As a result, the seat shell **200** may be better supported and stabilized on the base **300**. Furthermore, the specific shapes of the recessed portion **222** and the pedestal **322** of the projection **320** permit the seat shell **200** to be freely rotatable on the base **300**.

(64) The aperture **220** may be positioned within the recessed portion **222**. In particular, as may be best seen in FIG. **17**, the aperture **220** may be centrally positioned within the recessed portion **222**. As may be best seen in FIG. **17** with reference also to FIG. **13**, the aperture **222** in the lower portion **210** of the seat shell **200** may be shaped complementary to the post **324** and the flange(s) of the projection **320** of the base **300**. In this way, the post **324** and the flange(s) of the projection **320** of the base **300** may be at least partially received through the aperture **220**. Put another way, at least a portion of the post **324** and the flanges **326** of the projection **320** may be received through the aperture **220**. As a result, the seat shell **200** may be better supported and stabilized on the base **300**.

(65) The specific shapes of the aperture **220** and the post **324** and flange(s) **326** of the projection **320** permit the seat shell **200** to be removable or detachable from base **300**. In particular embodiments, the aperture **220** is oriented at an angle with respect to the first and second sides **202**, **204** of the seat shell **200** (in comparison to the projection **320** illustrated in FIG. **12** and FIG. **13**, which includes flanges **326** that extend outwardly away from the post **324** toward the first and second sides **302**, **304** of the base **300**). As a result, in such embodiments, the seat shell **200** is not removable or detachable from base **300** in either the forward-facing position or the rearward-facing position.

(66) With reference to FIG. **17** and FIG. **18**, the seat shell **200** may include one or more engagement features. For example, as shown in FIG. **17** and FIG. **18**, an upper engagement feature **230** may be positioned proximate the rear end **208** of the seat shell **200**. In particular, the upper engagement feature **230** may be positioned on the seat back **250** of the seat shell **200**. By way of non-limiting example, as may be best seen with reference to FIGS. **15A-B**, the upper engagement feature **230** may be formed as a hook. In particular, the upper engagement feature **230** may be formed as an upward-pointing or upward-oriented hook (i.e., in the direction of arrow **50** in FIGS. **15A-B**).

(67) A rear engagement feature **231** may be positioned proximate the rear end **208** of the seat shell **200**. In particular, the rear engagement feature **231** may be positioned proximate the location where the seat back **250** of the seat shell **200** meets the lower portion **210** of the seat shell **200**. More specifically, the rear engagement feature **231** may be positioned proximate the locking projection **238** on the lower portion **210** of the seat shell **200**. By way of non-limiting example, as may be best seen with reference to FIGS. **15A-B** and FIG. **19**, the rear engagement feature **231** may be formed as a hook. In particular, the rear engagement feature **231** may be formed as an upward-pointing or upward-oriented hook (i.e., in the direction of arrow **50** in FIGS. **15A-B**).

(68) A forward engagement feature **234** may be positioned proximate the forward end **206** of the seat shell **200**. By way of non-limiting example, as may be best seen with reference to FIGS. **15A-B** and FIG. **19**, the forward engagement feature **234** may be formed as a hook. In particular, the forward engagement feature **234** may be formed as an upward-pointing or upward-oriented hook (i.e., in the direction of arrow **50** in FIGS. **15A-B**).

(69) A first lower engagement feature **232** may be positioned proximate the rear end **208** of the seat shell **200**. In particular, the first lower engagement feature **232** may be positioned on the lower portion **210** of the seat shell **200** proximate the rear end **208** of the seat shell **200**. More specifically,

the first lower engagement feature **232** may be positioned between the rear engagement feature **231** and the aperture **220**. By way of non-limiting example, as may be best seen with reference to FIGS. **15A-B** and FIG. **20**, the first lower engagement feature **232** may be formed as a hook. In particular, the first lower engagement feature **232** may be formed as an upward-pointing or upward-oriented hook (i.e., in the direction of arrow **50** in FIGS. **15A-B**). It is to be understood that the first lower engagement feature **232** may be omitted in certain embodiments.

(70) A second lower engagement feature **233** may be positioned proximate the forward end **206** of the seat shell **200**. In particular, the second lower engagement feature **233** may be positioned on the lower portion **210** of the seat shell **200** proximate the forward end **206** of the seat shell **200**. More specifically, the second lower engagement feature **233** may be positioned between the forward engagement feature **234** and the aperture **220**. That is, the second lower engagement feature **233** may be positioned on an opposite side of the aperture **220** from the first lower engagement feature **232**. By way of non-limiting example, as may be best seen with reference to FIGS. **15A-B** and FIG. **19**, the second lower engagement feature **233** may be formed as a hook. In particular, the second lower engagement feature **233** may be formed as an upward-pointing or upward-oriented hook (i.e., in the direction of arrow **50** in FIGS. **15A-B**). It is to be understood that the second lower engagement feature **233** may be omitted in certain embodiments.

(71) In preferred embodiments, the engagement features of the seat shell **200** are shaped complementary to the receiving features of the base **300**. In this way, as described in detail herein, each of the receiving features of the base **300** is configured to engage a corresponding one of the engagement features of the seat shell **200** when the seat shell **200** is in the rearward-facing position or the forward-facing position. Engagement between the receiving features of the base **300** and corresponding ones of the engagement features of the seat shell **200** assists in providing structural integrity and stability to the car seat in certain situations, such as when the car seat undergoes significant movement or impact (e.g., during a car crash).

(72) With reference back to FIG. **3**, it should be understood that, when the seat shell **200** is in the forward-facing position: (a) the upper engagement feature **230** of the seat shell **200** engages with the upper receiving feature **330** of the base **300**; (b) the rear engagement feature **231** of the seat shell **200** engages with the intermediate receiving feature **331** of the base **300**; and (c) the first lower engagement feature **232** of the seat shell **200** engages with the lower receiving feature **332** of the base **300**. In this way, interaction between these engagement features **230**, **231**, **232** of the seat shell **200** and these receiving features **330**, **331**, **332** of the base **300** assist in retaining the seat shell **200** into the forward-facing position in certain situations, such as when the car seat undergoes significant movement or impact (e.g., during a car crash). As can be further understood, when the seat shell **200** is in the forward-facing position, the forward engagement feature **234** of the seat shell **200** and the second lower engagement feature **233** of the seat shell **200** do not engage the base **300**.

(73) Conversely, with reference back to FIG. **5**, it should be understood that, when the seat shell **200** is in the rearward-facing position: (a) the forward engagement feature **234** of the seat shell **200** engages with the intermediate receiving feature **331** of the base **300**; and (b) the second lower engagement feature **233** of the seat shell **200** engages with the lower receiving feature **332** of the base **300**. In this way, interaction between these engagement features **233**, **234** of the seat shell **200** and these receiving features **331**, **332** of the base **300** assist in retaining the seat shell **200** into the rearward-facing position in certain situations, such as when the car seat undergoes significant movement or impact (e.g., during a car crash). As can be further understood, when the seat shell **200** is in the rearward-facing position, the upper engagement feature **230** of the seat shell **200**, the rear engagement feature **231** of the seat shell **200**, and the first lower engagement feature **232** of the seat shell **200** do not engage the base **300**. Additionally, when the seat shell **200** is in the rearward-facing position, the upper receiving feature **330** of the base **300** does not engage the seat shell **200**.

(74) The rear and forward engagement features **231**, **234** of the seat shell **200** may be configured to

generally travel along the support ledge **358** of the base **300** as the seat shell **200** is rotated with respect to the base **300**. More specifically, the rear and forward engagement features **231**, **234** of the seat shell **200** may be configured to travel just above or on top of the support ledge **358** of the base **300** as the seat shell **200** is rotated with respect to the base **300**.

(75) As may be best seen in FIG. **17** and FIG. **18**, the rear engagement feature **231** of the seat shell **200** may include a detent or aperture **231a**. The forward engagement feature **234** of the seat shell **200** may also include a detent or aperture **234a**. The apertures **231a**, **234a** of the rear and forward engagement features **231**, **234** may be configured to engage with the base **300** as the seat shell **200** is rotated with respect to the base **300**. In particular, the apertures **231a**, **234a** of the rear and forward engagement features **231**, **234** may be configured to engage with the base **300** as the seat shell **200** is rotated with respect to the base **300** and the rear and forward engagement features **231**, **234** travel along the support ledge **358** of the base **300**. More specifically, each of the apertures **231a**, **234a** of the rear and forward engagement features **231**, **234** may be configured to engage with a corresponding one of the first and second retainment features **342**, **344** of the base **300** (e.g., on the support ledge **358** of the base **300**).

(76) When the seat shell **200** is in an intermediate position facing a first direction, the aperture **231a** of the rear engagement feature **231** engages the first retainment feature **342** of the base **300** and the aperture **234a** of the forward engagement feature **234** engages the second retainment feature **344** of the base **300**. When the seat shell **200** is rotated to an intermediate position facing an opposite, second direction, the aperture **234a** of the forward engagement feature **234** engages the first retainment feature **342** of the base **300** and the aperture **231a** of the rear engagement feature **231** engages the second retainment feature **344** of the base **300**. In this way, interaction between the apertures **231a**, **234a** of the rear and forward engagement features **231**, **234** of the seat shell **200** and the first and second retainment features **342**, **344** of the base **300** retain the seat shell **200** in the intermediate position(s). A user may then apply a relatively small amount of force to rotate the seat shell **200** out of the intermediate position(s). That is, actuation of the handle is not necessary (or even useful) to overcome the interaction between the engagement features of the seat shell **200** and the retainment features of the base **300** and rotate the seat shell **200** out of the intermediate position(s).

(77) Although the engagement features of the seat shell **200** are described herein as including a detent or aperture that engages with a corresponding retainment feature of the base **300** that is in the form of a projection or nub, it should be understood that these structures could be reversed. That is, the rear and forward engagement features of the seat shell **200** could be formed with projections or nubs and the retainment features of the base **300** could be formed as detents or apertures that engage the projections or nubs of the engagement features.

(78) With reference now to FIG. **18** and FIG. **19**, the lower portion **210** of the seat shell **200** may include one or more abutment features. In the exemplary embodiment illustrated in FIG. **18** and FIG. **19**, the lower portion **210** of the seat shell **200** includes a first abutment feature **203** positioned on the first side **202** of the seat shell **200** and a second abutment feature **205** positioned on the opposite, second side **204** of the seat shell **200**. More specifically, in this embodiment, the first and second abutment features **203**, **205** are each positioned proximate the forward end **206** of the seat shell **200**. The first and second abutment features **203**, **205** generally operate so as to prevent rotation of the seat shell **200** with respect to the base **300** when the seat shell **200** is in a reclined position (e.g., a partially or intermediate-reclined position). In particular, the first and second abutment features **203**, **205** may be configured to prevent rotation of the seat shell **200** with respect to the base **300** when the seat shell **200** is in any of a plurality of reclined positions (e.g., a partially or intermediate-reclined position). More specifically, the first and second abutment features **203**, **205** may be configured to prevent rotation of the seat shell **200** with respect to the base **300** when the seat shell **200** is any reclined position except for a fully-reclined position. That is, in certain embodiments, the seat shell **200** may be configured to freely and fully (i.e., a full 360°) rotate with

respect to the base **300** when the seat shell **200** is in the fully-reclined position. In particular embodiments, when the seat shell **200** is reclined with respect to the base **300** between the upright position and the fully-reclined position, such as in an intermediate-reclined position, the seat shell **200** may be prevented from freely and fully (i.e., a full 360°) rotating with respect to the base **300**. Rather, in such particular embodiments, in the intermediate-reclined position, the seat shell **200** may only be capable of rotating about 90° in either direction with respect to the base **300**. For example, in certain embodiments, when the seat shell **200** is in a partially reclined rearward-facing position (refer to FIG. 4 and FIG. 5), the seat shell may need to be placed in an upright position (e.g., as shown in FIG. 2 and FIG. 3 with respect to the forward-facing position) before the seat shell **200** can be fully rotated with respect to the base **300** into the forward-facing position. Placing the seat shell into the upright position before rotating the seat shell **200** with respect to the base **300** may help to ensure that the upper engagement feature **230** of the seat shell **200** is capable of engaging with the upper receiving feature **330** of the base **300**.

(79) Turning now to FIGS. 21-23, the seat shell **200** may include a recline mechanism **240**, which is also labeled in FIG. 16-20. FIG. 21 is a cross-sectional view taken along line 21-21' of FIG. 16 and showing features and details of the recline mechanism **240**, and FIG. 22 and FIG. 23 are first and second side cutaway views, respectively, in which the outer surface **216** of the lower portion **210** of the seat shell **200** has been removed to show additional features and details of the recline mechanism **240**. The recline mechanism **240** may generally operate so as to permit the seat shell **200** to be reclinable with respect to the base **300**. In particular, the seat shell **200** may travel (e.g., slidably or otherwise) along a reclination path (e.g., an arcuate reclination path) defined by the recline mechanism **240**. In this way, the complementary shapes of the lower portion **210** of the seat shell **200** and the receiving portion **310** of the base **300** may advantageously assist in smooth travel of the seat shell **200** as the seat shell **200** is reclined with respect to the base **300**. In particular embodiments, the recline mechanism **240** may assist in ensuring that a substantially constant gap is maintained between the lower portion **210** of the seat shell **200** and the receiving portion **310** of the base **300** such that the interface between the seat shell **200** and the base **300** is maintained as the seat shell **200** is rotated and/or reclined with respect to the base **300**, including as the seat shell **200** is rotated with respect to the base **300** while in a reclined position or upright position. This advantageously allows the seat shell **200** to be reclined and/or rotated with respect to the base **300** without any need to remove or detach the seat shell **200** from the base **300** or to uninstall or reinstall the seat shell **200** onto the base **300**, including allowing the seat shell **200** to be rotated with respect to the base **300** while in a reclined position or upright position.

(80) The recline mechanism **240** may be formed as a part of the seat bottom **209** of the seat shell **200**. In particular, the recline mechanism **240** may be formed as a part of the lower portion **210** of the seat bottom **209** of the seat shell **200**. The recline mechanism **240** may, in certain constructions, be integrally formed with the seat shell **200**. In certain constructions, the recline mechanism **240** may include the aperture **220**, as may be best seen in FIG. 22. In this way, the recline mechanism **240** may be supported on the base **300** (e.g., on the projection **320** in the receiving portion **310** of the base **300**). The recline mechanism **240** may therefore be at least partially disposed within the receiving portion **310** of the base **300** when the seat shell **200** is attached to the base **300**.

(81) As may be best seen in FIG. 21 and FIG. 22, the recline mechanism **240** may be arcuate. Put another way, the recline mechanism **240** may generally be shaped similarly to the lower portion **210** of the seat shell **200** and may be convex. At least some of the engagement features of the seat shell **200** may connect to and extend outwardly away from the recline mechanism **240**. The recline mechanism **240** may generally extend from proximate the forward end **206** of the seat shell **200** to proximate the rear end **208** of the seat shell. The recline mechanism **240** may generally be positioned approximately centrally between the first side **202** and the second side **204** of the seat shell **200**. In certain constructions, the recline mechanism **240** may be positioned between the first and second abutment features **203**, **205**. In embodiments, the first and second abutment features

203, 205 are not directly connected to or positioned on the recline mechanism **240**.

(82) The recline mechanism **240** generally includes at least one track. The at least one track is generally formed as an elongate track such that a mechanism inserted therein or therethrough may generally travel along the elongate track. In the exemplary embodiment illustrated in FIG. **21** and FIG. **22**, the recline mechanism **240** includes four tracks. In the illustrated exemplary embodiment, the tracks are defined in first and second arcuate bars **244a, 244b** that are spaced apart from one another. In particular, each of the first and second bars **244a, 244b** define a pair of tracks. For example, first bar **244a** defines track **242a** proximate the forward end **206** of the seat shell **200** and track **242c** proximate the rear end **208** of the seat shell **200**, as may be best seen in FIG. **22** and FIG. **23**. Similarly, second bar **244b** defines track **242b** proximate the forward end **206** of the seat shell **200** and track **242d** proximate the rear end **208** of the seat shell **200**. In this exemplary embodiment, tracks **242a** and **242c** are positioned proximate the first side **202** of the seat shell **200**, and tracks **242b** and **242d** are positioned proximate the second side **204** of the seat shell **200**. Moreover, tracks **242a** and **242b** are generally positioned directly across from one another (e.g., equidistantly spaced between the forward and rear ends **206, 208** of the seat shell **200**), and tracks **242c** and **242d** are likewise generally positioned directly across from one another (e.g., equidistantly spaced between the forward and rear ends **206, 208** of the seat shell **200**). In this way, rod **246a** generally extends through tracks **242a** and **242b**, and rod **246b** generally extends through tracks **242c** and **242d**.

(83) The track(s) of the recline mechanism **240** is configured to receive a rod therethrough. The rod is adapted for movement within the track as the seat shell **200** is reclined with respect to the base **300**. For example, with reference again to FIG. **21** and FIG. **22**, tracks **242a** and **242b** receive rod **246a** therethrough, and tracks **242c** and **242d** receive rod **246b** therethrough. Rod **246a** is adapted for movement within tracks **242a** and **242b** (i.e., by moving along tracks **242a** and **242b** toward and away from the forward and rear ends **206, 208** of the seat shell **200**). Similarly, rod **246b** is adapted for movement within tracks **242c** and **242d** (i.e., by moving along tracks **242c** and **242d** toward and away from the forward and rear ends **206, 208** of the seat shell **200**). The rod(s) moving within the track(s) advantageously allows the seat shell **200** to be reclined with respect to the base **300** without any need to remove or detach the seat shell **200** from the base **300** or to uninstall or reinstall the seat shell **200** onto the base **300**.

(84) The recline mechanism **240** generally defines a plurality of openings **248**. Each of the plurality of openings **248** generally operates such that a mechanism inserted therein or therethrough is retained therein. In particular, each opening **248** or pair of openings **248** may define a discrete recline position for the seat shell **200** (e.g., an upright position, a fully-reclined position, an intermediate-reclined position). For example, in the exemplary embodiment illustrated in FIG. **21** and FIG. **22**, eight openings are defined, which corresponds to four discrete recline positions (e.g., an upright position, a fully-reclined position, and two intermediate-reclined positions). Generally, the one or more openings (e.g., pair of openings) defined at one of the extreme ends of the openings corresponds to and defines an upright position, and the one or more openings (e.g., pair of openings) defined at the opposite extreme end corresponds to and defines a fully-reclined position. The remaining openings (e.g., pairs of openings) disposed between the extreme ends correspond to and define intermediate-reclined positions. Each of the intermediate-reclined positions and the fully-reclined position may all be considered reclined positions.

(85) In the exemplary embodiment illustrated in FIG. **21** and FIG. **22**, the recline mechanism **240** includes a first set of openings **248a** and a second set of openings **248b**, as may be best seen in FIG. **22**. In the illustrated exemplary embodiment, the first and second sets of openings **248a, 248b** are defined in the first and second arcuate bars **244a, 244b**, respectively. In particular, each of the first and second bars **244a, 244b** define a set of openings. For example, first bar **244a** defines the first set of openings **248a**, with individual ones of the openings being spaced apart from one another between the forward end **206** of the seat shell **200** the rear end **208** of the seat shell **200**.

Similarly, second bar **244b** defines the second set of openings **248b**, with individual ones of the openings being spaced apart from one another between the forward end **206** of the seat shell **200** and the rear end **208** of the seat shell **200**.

(86) The openings **248** of the recline mechanism **240** are configured to receive a tube therethrough. For example, with reference again to FIG. **22** and FIG. **23**, openings **248a'** and **248a''** receive tube **249a** therethrough, and openings **248b'** and **248b''** receive tube **249b** therethrough. Tubes **249a** and **249b** are each adapted to retain the seat shell **200** in a selected one of the reclined positions as determined by which opening(s) into which the tube is inserted. The tube(s) of the recline mechanism **240** may generally be operatively connected to an actuator (not shown) that is selectively controlled by a user to disengage the tube from an opening of the recline mechanism **240** so as to recline the seat shell **200** with respect to the base **300**. In this way, the amount of recline of the seat shell **200** with respect to the base **300** can be selectively controlled, and the seat shell **200** can be selectively reclinable into a plurality of positions (e.g., an upright position, a fully-reclined position, one or more intermediate-reclined positions).

(87) While particular exemplary embodiments have been described, it is not intended that the claims be limited thereto, as it is intended that the claims be as broad in scope as the art will allow and that the specification be read likewise. That is, the foregoing description of specific embodiments will so fully reveal the general nature of the disclosure that others may, by applying knowledge within the skill of the art, readily modify and/or adapt for various applications such specific embodiments, without undue experimentation, without departing from the general concept of the present disclosure. Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed embodiments, based on the teaching and guidance presented herein. It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by the skilled artisan in light of the teachings and guidance.

(88) It will be appreciated by those of ordinary skill in the art that the components, method steps and materials illustrated above may be varied by substitution of equivalent components, steps and materials capable of performing the same functions. It will also be appreciated by one of ordinary skill in the art that sizes and strengths of the components may be scaled up or down as required for specific purposes. The claims hereof are intended to encompass all such equivalent components, method steps and scales.

Claims

1. A car seat, comprising: a base including a receiving portion, wherein the receiving portion comprises a concave surface that defines a bowl shape; a projection extending upwardly away from an inner surface of the receiving portion of the base, wherein the projection includes a pedestal, a post extending upwardly away from the pedestal, and a flange extending outwardly away from the post, wherein the flange is spaced apart from the pedestal along a length of the post; a seat shell including a seat bottom having a lower portion forming an outer convex surface that at least partially defines a dome shape, the seat bottom lower portion is configured to interface with the receiving portion of the base, wherein the seat shell is rotatable with respect to the base about the projection such that the seat shell is selectively positionable between at least a rearward-facing position and a forward-facing position; and an aperture in the lower portion of the seat shell and shaped complementary to the post and the flange, wherein the flange and at least a portion of the post can be received through the aperture such that a shape of the aperture, post, and flange permit the seat shell to be removeable from the base.

2. The car seat of claim 1, wherein: the lower portion of the seat shell includes a recessed portion shaped complementary to the pedestal of the projection such that the pedestal of the projection is at

least partially received in the recessed portion; and the aperture in the lower portion of the seat shell is centrally positioned within the recessed portion.

3. A car seat, comprising: a base including a receiving portion having a projection extending from an inner surface of the receiving portion of the base, wherein the projection includes a pedestal, a post extending from the pedestal, and a flange extending outwardly away from the post, wherein the flange is spaced apart from the pedestal along a length of the post; and a seat shell including a seat bottom having a lower portion forming, at least in part, a convex surface, the seat bottom lower portion is configured to interface with the receiving portion of the base, wherein the seat shell is rotatable with respect to the base such that the seat shell is selectively positionable between at least a rearward-facing position and a forward-facing position, wherein the seat shell comprises an aperture in the lower portion of the seat shell and shaped complementary to the post and the flange, wherein the flange and at least a portion of the post can be received through the aperture.

4. The car seat of claim 3, wherein the receiving portion is, at least in part, comprised of a concave surface.

5. The car seat of claim 4, wherein the convex surface of the lower portion of the seat bottom is complimentary in shape to the concave surface of the receiving portion.

6. The car seat of claim 4, wherein: the convex surface of the lower portion at least partially defines a dome shape; and the concave surface of the receiving portion at least partially defines a bowl shape.

7. The car seat of claim 3, wherein: the lower portion of the seat shell includes a recessed portion shaped complementary to the pedestal of the projection such that the pedestal of the projection is at least partially received in the recessed portion; and the aperture in the lower portion of the seat shell is centrally positioned within the recessed portion and is shaped complementary to the post and at least a portion of the post is received through the aperture.

8. The car seat of claim 3, wherein the pedestal is shaped so as to at least partially define a cone shape extending upwardly away from the inner surface of the receiving portion of the base.

9. The car seat of claim 3, wherein the base is comprised of a lock configured to lock rotation of the seat shell with respect to the base in each of the rearward-facing position and the forward-facing position and the car seat further comprises a first indicator on a first side of the base, the first indicator configured to indicate whether rotation of the seat shell is locked with respect to the base.

10. The car seat of claim 3, wherein the base is comprised of a lock configured to lock rotation of the seat shell with respect to the base in each of the rearward-facing position and the forward-facing position and the car seat further comprises at least one indicator at the seat shell, the at least one indicator of the seat shell configured to indicate whether rotation of the seat shell is locked with respect to the base.

11. The car seat of claim 3, wherein the base includes: a backing portion extending upwardly away from the receiving portion; and an upper tethering point proximate an upper edge of the backing portion of the base, the upper tethering point configured to remain stationary as the seat shell is rotated with respect to the base.

12. The car seat of claim 3, wherein: the seat shell includes one or more engagement features; and the base includes one or more receiving features, wherein each of the one or more receiving features is configured to engage a corresponding one of the one or more engagement features when the seat shell is in at least one of the rearward-facing position and the forward-facing position.

13. The car seat of claim 12, wherein: the one or more receiving features includes a first receiving feature and a second receiving feature and the one or more engagement features includes a first engagement feature and a second engagement feature; the first receiving feature is positioned within the receiving portion of the base and is configured to engage the first engagement feature positioned on the lower portion of the seat shell; and the second receiving feature is positioned adjacent the receiving portion of the base and is configured to engage the second engagement feature.

14. The car seat of claim 3, wherein the receiving portion further comprises a support ledge that is positioned radially inward of an outer edge of the receiving portion and wherein the support ledge is circular in shape.
15. The car seat of claim 3, wherein the base further comprises a belt lock off arm having a proximal end moveably connected to the base, wherein the belt lock off arm can move between a locked position and an unlocked position.
16. The car seat of claim 15, wherein the base further comprises a recessed portion of the receiving portion, wherein the recessed portion is capable of receiving the belt lock off arm in the locked position in a manner that allows rotation of the seat shell in the base from the rearward-facing position to the forward-facing position.
17. The car seat of claim 3 further comprising a recline mechanism between a top surface of the seat bottom and the seat bottom lower portion, the recline mechanism configured to permit the seat shell to recline with respect to the base such that the seat shell is selectively reclinable between at least an upright position and at least one reclined position.
18. A car seat, comprising: a base including a receiving portion having a pedestal, a post extending away from the pedestal, and a flange extending away from the post, wherein the flange is spaced apart from the pedestal along a length of the post; a seat shell including a seat bottom having a top surface and a lower portion forming a convex surface, the seat bottom lower portion interfacing with the receiving portion of the base, wherein the seat shell is rotatable about the post with respect to the base such that the seat shell is selectively positionable between at least a rearward-facing position and a forward-facing position, wherein the seat shell comprises an aperture in the lower portion of the seat shell and shaped complementary to the post and the flange, wherein the flange and at least a portion of the post can be received through the aperture; and a recline mechanism between the seat bottom top surface and the seat bottom lower portion, the recline mechanism configured to permit the seat shell to recline with respect to the base such that the seat shell is selectively reclinable between at least an upright position and at least one reclined position.
19. The car seat of claim 18, wherein: the seat shell includes one or more engagement features; and the base includes one or more receiving features; wherein each of the one or more receiving features is configured to engage a corresponding one of the one or more engagement features when the seat shell is in at least one of the rearward-facing position or the forward-facing position.
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